Curriculum Vitae of Chongzhi YIN

Chongzhi YIN (殷翀之), Ph.D.

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Research Experience & Appointments

- **Specially Appointed Assistant Professor**
  - *Mar. 2026 – Present*
  - Graduate School of Information Science, University of Hyogo, Kobe, Japan
  - **Supervisor:** Prof. Shin-ichiro Shima
  - **Project:** UAE Research Program for Rain Enhancement Science (UAEREP)
  - **Research:** Develop and evaluate SCALE-SDM simulations of UAE convective clouds using ERA5-derived soundings and satellite cloud diagnostics.
  - **Rain enhancement:** Design and analyze idealized sensitivity experiments for prescribed collision–coalescence enhancement, application timing and eligible droplet sizes.

Education

- **Ph.D. in Atmospheric Physics and Atmospheric Environment**
  - *Sep. 2019 – Jun. 2025*
  - School of Atmospheric Physics, Nanjing University of Information Science & Technology (NUIST), Nanjing, China
  - **Dissertation:** A Simulation Study on the Numerical Convergence of the Super-Droplet Method and the Macro- and Micro-physical Characteristics of Stratocumulus Clouds
  - **Supervisor:** Prof. Chunsong Lu
  - **Co-supervisor:** Prof. Shin-ichiro Shima
- **Research Student**
  - *Nov. 2020 – Dec. 2022*
  - Graduate School of Information Science, University of Hyogo, Kobe, Japan
  - **Research Focus:** Development and application of Lagrangian cloud microphysics schemes.
  - **Supervisor:** Prof. Shin-ichiro Shima
- **M.S. phase, Combined M.S.-Ph.D. Program in Atmospheric Physics**
  - *Sep. 2017 – Jun. 2019*
  - School of Atmospheric Sciences, NUIST, Nanjing, China
  - **Supervisor:** Prof. Tianliang Zhao and Assoc. Prof. Qian Huang

- **B.S. in Atmospheric Sciences (Sino-foreign Cooperation Program)**
  - *Sep. 2012 – Jun. 2017*
  - School of Atmospheric Physics, NUIST, Nanjing, China
  - **Undergraduate Thesis Advisor:** Assoc. Prof. Qian Huang

Research Interests

- Cloud Microphysics and Dynamics
- Large Eddy Simulation (LES) & Direct Numerical Simulation (DNS)
- Particle-based Models (Super-Droplet Method, SDM)
- Lagrangian Particle Tracking, Two-Pass Hybrid Tracking (TPHT) & Trajectory Analysis
- Turbulence-Microphysics Interactions in Stratocumulus and Convective Clouds
- Development of Cloud Microphysics Parameterizations

Honors & Awards

- **China's Top 10 Advances in Meteorological Science and Technology**, 2024
  - *Awarded for the project "Improvement of cloud microphysics schemes and their application in weather and climate models". (Team Member, 13th Contributor)*
- **Excellent Oral Presentation Award**, 7th Chinese Symposium on Mesoscale Meteorology, 2025
- **Excellent Oral Presentation Award**, 1st IAMES Conference, 2021
- **China Scholarship Council (CSC) Scholarship**, State-Sponsored Graduate Program, Aug. 2020
  - *Awarded for joint Ph.D. study at the University of Hyogo, Japan.*

Publications

Peer-Reviewed Journal Articles *(* denotes first author)*

1. ***Yin, C.****, Shima, S.-I., & Lu, C. (2026). A Lagrangian particle tracking framework for the super-droplet method in SCALE-SDM 5.2.6-2.3.1: implementation and computational evaluation. *Geoscientific Model Development*, 19(20), 9993–10022. https://doi.org/10.5194/gmd-19-9993-2026.
2. ***Yin, C.****, Shima, S., Lu, C., Gao, S., & Xu, X. (2026). Resolving Entrainment–Mixing in Marine Stratocumulus: The Role of LES Grid Resolution and Super-Droplet Number. *Advances in Atmospheric Sciences*, 43(4), 845–860. https://doi.org/10.1007/s00376-025-5043-z.
3. Yang, F., Choi, K. O., Chandrakar, K. K., Hoffmann, F., Hou, P., Krueger, S., Lu, C., Ovchinnikov, M., Ren, Y., Shima, S., Wu, P., **Yin, C.**, Zhu, Z., & Yum, S. S. (2026). A Model Intercomparison Study to Investigate Mixing Characteristics in Non-Precipitating Stratocumulus Clouds. *Journal of Advances in Modeling Earth Systems*, 18(4), e2025MS005620. https://doi.org/10.1029/2025MS005620.
4. ***Yin, C.****, Shima, S., Xue, L., Lu, C., et al. (2024). Simulation of marine stratocumulus using the super-droplet method: numerical convergence and comparison to a double-moment bulk scheme using SCALE-SDM 5.2.6-2.3.1. *Geoscientific Model Development*, 17(13), 5167-5189. https://doi.org/10.5194/gmd-17-5167-2024.

5. Jia, M., Cheng, X., Zhao, T., **Yin, C.**, et al. (2019). Regional Air Quality Forecast Using a Machine Learning Method and the WRF Model over the Yangtze River Delta, East China. *Aerosol and Air Quality Research*, 19, 1602-1613. https://doi.org/10.4209/aaqr.2019.05.0275.
6. Zhang, Y., Zhao, T., **Yin, C.**, et al. (2019). Seasonal variation of the relationship between surface PM2.5 and O3 concentrations in Xuzhou. *China Environmental Science*, 39(6), 2267-72. (In Chinese)

Research Software

- **Yin, C.** (2026). *SCALE-SDM super-droplet tracking suite with forward/backward tracking and TPHT* (Version GMD2026_FW_BW_TPHT_v1.2.1) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.20659293.
    - **Contribution:** Co-developed forward, backward and TPHT algorithms; performed numerical experiments, computational evaluation, post-processing and analysis, and prepared the first-author manuscript.

Teaching & Professional Activities

- **Instructor, 2nd SCALE-SDM Hands-on Tutorial**
    - University of Hyogo, Kobe, Japan, 26–27 October 2026. Certificate of Appreciation issued 27 October 2026 (verify credential).
- **Reviewer for Peer-Reviewed Journals**
    - Journal of Geophysical Research: Atmospheres (2023, 2024)
    - Atmospheric Chemistry and Physics (2026)

Research Projects

- **Participant**, UAE Research Program for Rain Enhancement Science (UAEREP), University of Hyogo, 2026–Present. SCALE-SDM modeling of UAE convective clouds and idealized rain-enhancement sensitivity experiments.
- **Participant**, Project: "Advancing the SCALE Weather and Climate Modeling Framework through Generative-AI Spec-Driven Development", Next-Generation Computational Science Grand Reach Program (Category B), HPCI project hp260519. Allocation period: 24 September 2026–31 March 2027.
- **Participant**, Project: "From Validation to Application: Advancing the Super-Droplet Method and Evaluating Weather Intervention Strategies for Extreme Rainfall Controllability in Japan", Fugaku General Access, FY2026 (Period A), HPCI project hp260140. Allocation period: 1 April 2026–31 March 2027.
- **Participant**, Fugaku Start-up Project (rs240006), HPCI. Allocation period: 1 April 2024–31 March 2031.
- **Participant**, Project: "Validation and sophistication of the super-droplet method towards real-world weather simulation", HPCI General Access resource allocations, Japan, FY2020–FY2026 (1 April 2020–31 March 2027). Project IDs: hp200078, hp210059, hp220062, hp230166, hp240151, hp250136, hp260163.
- **Participant**, Project: "Numerical convergence characteristics of marine stratocumulus simulated by the super-droplet method", Funded by Earth System Numerical Simulation Facility (EarthLab), 2023.

- **Participant**, Project: "Numerical simulation study on the activation effect of aerosols on deep convection", Funded by Earth System Numerical Simulation Facility (EarthLab), 2022.

Conference Presentations & Reports

- **Chongzhi Yin** (2026, September 17). *Cloud-Particle History Reconstruction with Two-Pass Hybrid Tracking (TPHT) in SCALE-SDM and a Preliminary UAEREP Study of Idealized Laser Application for Rain Enhancement*. MS8 Cross-Project Workshop on Weather Prediction and Control, The University of Tokyo, Hongo Campus, Tokyo, Japan (Oral).
- **Chongzhi Yin** (2025, November 7). *Accurate Cloud Modeling using the Super-Droplet Method (in Chinese)*. Invited talk at Lanzhou University, Lanzhou, China (Online oral).
- **Chongzhi Yin**, Shin-ichiro Shima, Chunsong Lu (2025, October 22). *A Large-Eddy Simulation Study of Stratocumulus Using Lagrangian Super-Droplet Tracking (in Chinese)*. The 36th Annual Meeting of the Chinese Meteorological Society, Nanjing, China (Oral).
- **Chongzhi Yin**, Shin-ichiro Shima, Chunsong Lu (2025, September 26). *A Lagrangian Particle Tracking Framework for the Super-Droplet Method: Development, Implementation, and Application of Backward and Forward Algorithms in SCALE-SDM*. The Mini-workshop on particle-based cloud modeling 2025, Kobe, Japan (Invited).
- **Chongzhi Yin**, Shin-ichiro Shima, Chunsong Lu (2025, August). *Development and Application of a Super-Droplet Particle Tracking Algorithm in Large Eddy Simulations (in Chinese)*. The 7th Chinese Symposium on Mesoscale Meteorology, Datong, China (Oral).
- Fan Yang, KyoungOck Choi, Kamal Kant Chandrakar, Fabian Hoffmann, Pei Hou, Steve Krueger, Chunsong Lu, Mikhail Ovchinnikov, Yangze Ren, Shin-ichiro Shima, Peng Wu, **Chongzhi Yin**, Seong Soo Yum, Zeen Zhu (2025, July 20–25), *Entrainment-mixing characteristics for non-precipitating stratocumulus clouds: a model intercomparison study*, BACO 2025, Busan IAMAS-IACS-IAPSO Joint Assembly, BEXCO, Busan, Republic of Korea (Poster).
- Fan Yang, KyoungOck Choi, Kamal Kant Chandrakar, Fabian Hoffmann, Pei Hou, Steve Krueger, Chunsong Lu, Mikhail Ovchinnikov, Yangze Ren, Shin-ichiro Shima, Peng Wu, **Chongzhi Yin**, Seong Soo Yum, Zeen Zhu (2025, March 3–6), *Mixing characteristics in non-precipitation stratocumulus clouds: a model intercomparison study*, 2025 Joint ARM User Facility and ASR PI Meeting, Rockville Hilton Hotel and Executive Meeting Center in Rockville, Maryland, and online (Poster).
- **Chongzhi Yin**, Shin-ichiro Shima, Chunsong Lu (2024, July). *Preliminary Results from the ACE-ENA Case and Latest Findings from DYCOMSII-RF02 case*. The 11th International Cloud Modeling Workshop (ICMW), Seoul, South Korea (Oral).
- **Chongzhi Yin**, Shin-ichiro Shima, Chunsong Lu, Lulin Xue. (2023, January). *Simulation of Marine Stratocumulus Using the Super-Droplet Method*. The 15th Symposium on Aerosol-Cloud-Climate Interactions, AMS 103rd Annual Meeting, Denver, CO (Online oral).
- **Chongzhi Yin**, Shin-ichiro Shima, Chunsong Lu, Lulin Xue. (2022, December). *Simulation of Marine Stratocumulus Using the Super-Droplet Method*. The AGU Fall Meeting 2022, Chicago, IL (Online poster).
- **Chongzhi Yin**, Shin-ichiro Shima, Chunsong Lu, Lulin Xue. (2022, March). *Simulation of Drizzling Marine Stratocumulus Using the Super-droplet Method*. The 4th International Workshop on Cloud Turbulence, Nagoya, Japan (Online, invited).

- **Chongzhi Yin**, Shin-ichiro Shima, Chunsong Lu, Lulin Xue. (2021, November). *Simulation of Drizzling Marine Stratocumulus Using the Super-droplet Method*. The 1st IAMES Conference, Nanjing, China (Online oral).
- **Chongzhi Yin**, Shin-ichiro Shima, Chunsong Lu, Lulin Xue. (2021, August). *Simulation of Drizzling Marine Stratocumulus Using the Super-droplet Method*. The AOGS2021 Conference (Online oral).
- **Chongzhi Yin,** Shin-ichiro Shima, Chunsong Lu, Lulin Xue. (2021, February 16-17). *Numerical Convergence study on A Drizzling Marine Stratocumulus Case*. The Workshop on Aerosol-Cloud-Precipitation Interactions, Japan (Online oral).
- **Chongzhi Yin,** Tianliang Zhao, Xiangzhi Zhang, Mengwei Jia, Yujing Zhang. (2019, July). *Short-term Statistical Forecast and Assessment of PM2.5 Concentration in the Yangtze River Delta Region Based on a BP Artificial Neural Networks*. The 18th CTWF International Symposium on Aerosol and Climate Change: Observations, Modeling & Interactions, Beijing, China (Poster).
- **Chongzhi Yin,** Tianliang Zhao, Xiangzhi Zhang, Mengwei Jia, Yujing Zhang. (2018, November 3). *Short-term Statistical Forecast and Evaluation of PM2.5 Concentration in the Yangtze River Delta Region Based on a BP Neural Network Method (in Chinese)*. The 2018 Annual Academic Meeting of the Atmospheric Environment Branch, Chinese Society for Environmental Sciences (CSES), Qingdao, China (Oral).

Technical Skills

- **Modeling**: Large Eddy Simulation (LES), Super-Droplet Method (SDM), SCALE-SDM, WRF.
- **Programming & Scripting**: Fortran, Python, MATLAB, Bash Scripting.
- **High-Performance Computing (HPC)**: Parallel Computing (MPI, OpenMP), batch scheduling and restart diagnostics on SQUID.
- **Data Analysis & Visualization**: ParaView, Matplotlib, NCL.
- **Operating Systems & Tools**: Linux, macOS, Git, Obsidian.
- **Languages**: Chinese (Native), English (Fluent in academic communication).